

A large, faint watermark of the ISTAD logo is centered in the background. It is a circular emblem with a blue outer ring containing the text 'INSTITUTE OF SCIENCE AND TECHNOLOGY ADVANCED DEVELOPMENT' and Khmer text. Inside the ring is a blue circle with binary code '01010011 01010100 01000001 0100100' around the perimeter. The center features a stylized atom with a blue nucleus and three white orbits.

DevOps

Domain Name & Web Server

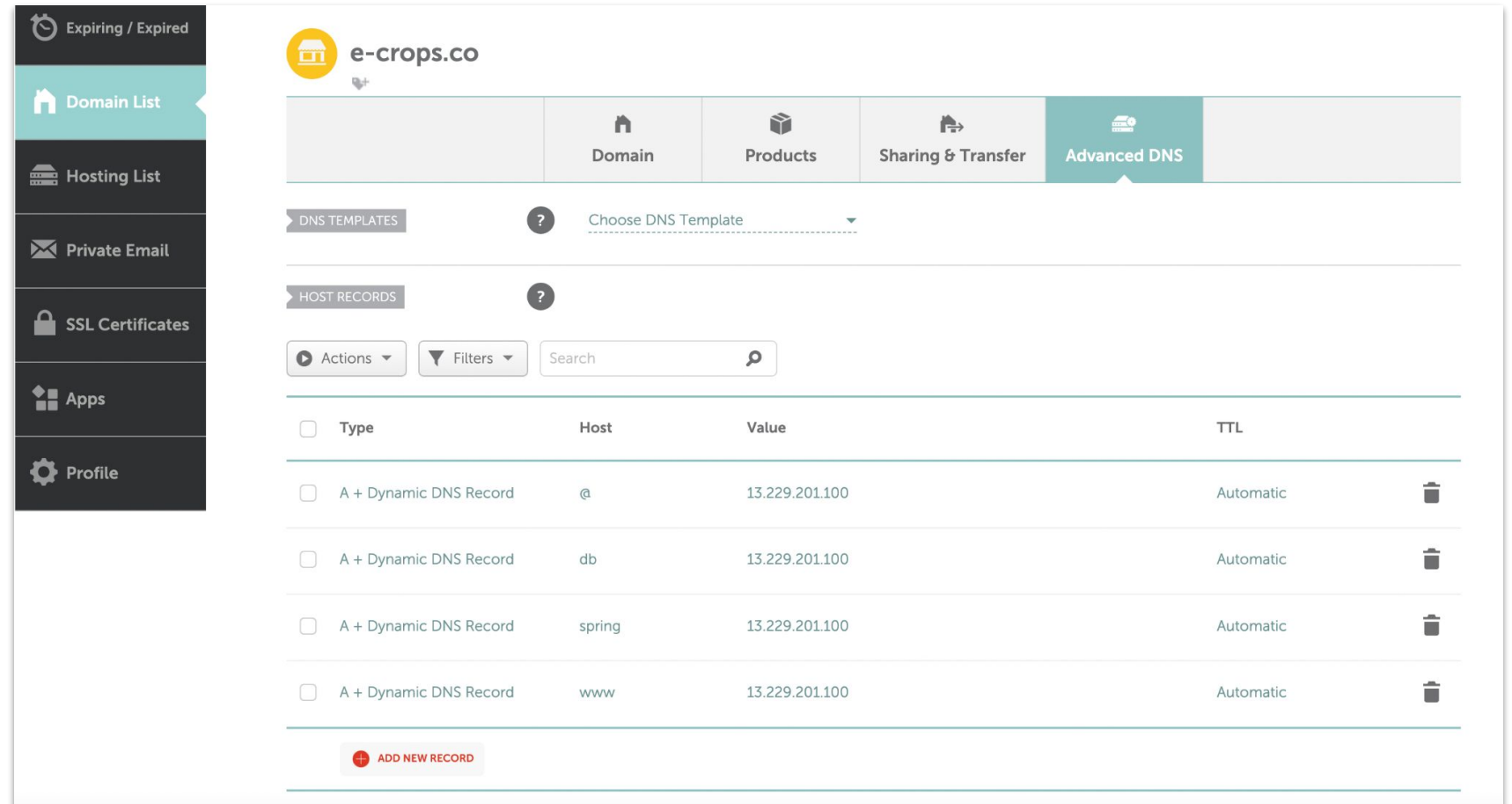
What is Domain Name?

- **A domain name** is a string of text that maps to a numeric IP address, used to access a website from client software. **In plain English**, a domain name is the text that a user types into a browser window to reach a particular website. For instance, the domain name for Google is '**google.com**'.
- The actual address of a website is a complex numerical IP address (e.g. 103.21.244.0), but thanks to **DNS**, users are able to enter human-friendly domain names and be routed to the websites they are looking for. This process is known as a DNS lookup.

What is Domain Name?

- We can easily purchased the domain name from web hosting provider such as:

- GoDaddy
- Namecheap
- blueHost
- HostGator
- Hostinger
- Web.com ...



The screenshot displays the domain management interface for e-crops.co. The left sidebar contains navigation options: Expiring / Expired, Domain List (selected), Hosting List, Private Email, SSL Certificates, Apps, and Profile. The main content area shows the domain name and a navigation bar with tabs: Domain, Products, Sharing & Transfer, and Advanced DNS (selected). Below the navigation bar, there are sections for DNS TEMPLATES and HOST RECORDS. The HOST RECORDS section includes a table with columns: Type, Host, Value, and TTL. The table lists four A + Dynamic DNS Records for the hosts @, db, spring, and www, all pointing to the value 13.229.201.100 with an Automatic TTL. An 'ADD NEW RECORD' button is located at the bottom of the table.

Type	Host	Value	TTL
☐ A + Dynamic DNS Record	@	13.229.201.100	Automatic
☐ A + Dynamic DNS Record	db	13.229.201.100	Automatic
☐ A + Dynamic DNS Record	spring	13.229.201.100	Automatic
☐ A + Dynamic DNS Record	www	13.229.201.100	Automatic

What is Domain Name?

- Domains are the **unique names** that identify an Internet site. A domain name must be unique so that Internet users can find the site. In order to set up domain name we need web hosting server to host our website. Nginx is the popular to help us.
- **Nginx** is a web server that is used to host websites. It is a free and open source software that is available for Windows, Linux, and OS X. Nginx is a very popular web server and is used by many large companies, such as Facebook, Netflix, and WordPress.
- Nginx is a good choice for websites that receive a lot of traffic.

What is Web Server?

- A **web server** is software and hardware that uses HTTP (Hypertext Transfer Protocol) and other protocols to respond to client requests made over the World Wide Web. The main job of a web server is to display website content through storing, processing and delivering web pages to users. Besides HTTP, web servers also support SMTP (Simple Mail Transfer Protocol) and FTP (File Transfer Protocol), used for email, file transfer and storage.



Nginx Web Server

- **Nginx** is an open source, fast, lightweight and high-performance web server that can be used to serve static files. NGINX is considered as the popular web server behind Apache web server and Microsoft's IIS.
- Features of NGINX
 - Reverse proxy with caching
 - IPV6
 - Load Balancing
 - Web Sockets
 - Handling of static files, index files, and auto-indexing
 - FastCGI support with caching
 - URL rewriting and redirection



Nginx Web Server

- `$ sudo apt install nginx -y`
- `$ sudo systemctl start nginx && sudo systemctl enable nginx`
- `$ cd /etc/nginx/conf.d`
- `$ sudo vim server.conf`

.conf match included file in /etc/nginx/nginx.conf

```
server {  
    listen 80;  
    listen [::]:80;  
    root /var/www/html;  
    index index.html index.htm index.nginx-debian.html;  
    server_name e-crops.co www.e-crops.co;  
    location / {  
        try_files $uri $uri/ =404;  
    }  
}
```

Nginx Web Server

- **Host:** This header contains the original host requested by the client, which is the website domain and port. Nginx keeps this in the `$http_host` variable.
- **X-Forwarded-For:** This header contains the IP address of the client who sent the original request. It can also contain a list of IP addresses, with the original client IP coming first, then a list of all the IP addresses of the reverse proxy servers that passed the request through. Nginx keeps this in the `$proxy_add_x_forwarded_for` variable.
- **X-Real-IP:** This header always contains a single IP address that belongs to the remote client. This is in contrast to the similar X-Forwarded-For which can contain a list of addresses. If X-Forwarded-For is not present, it will be the same as X-Real-IP.
- **X-Forwarded-Proto:** This header contains the protocol used by the original client to connect, whether it be HTTP or HTTPS. Nginx keeps this in the `$scheme` variable.

Nginx Web Server

```
server {  
    listen 80;  
    listen [::]:80;  
    root /var/www/html;  
    index index.html index.htm index.nginx-debian.html;  
    server_name e-crops.co www.e-crops.co;  
    location / {  
        proxy_set_header Host $http_host;  
        proxy_set_header X-Real-IP $remote_addr;  
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;  
        proxy_set_header X-Forwarded-Proto $scheme;  
  
        try_files $uri $uri/ =404;  
    }  
}
```

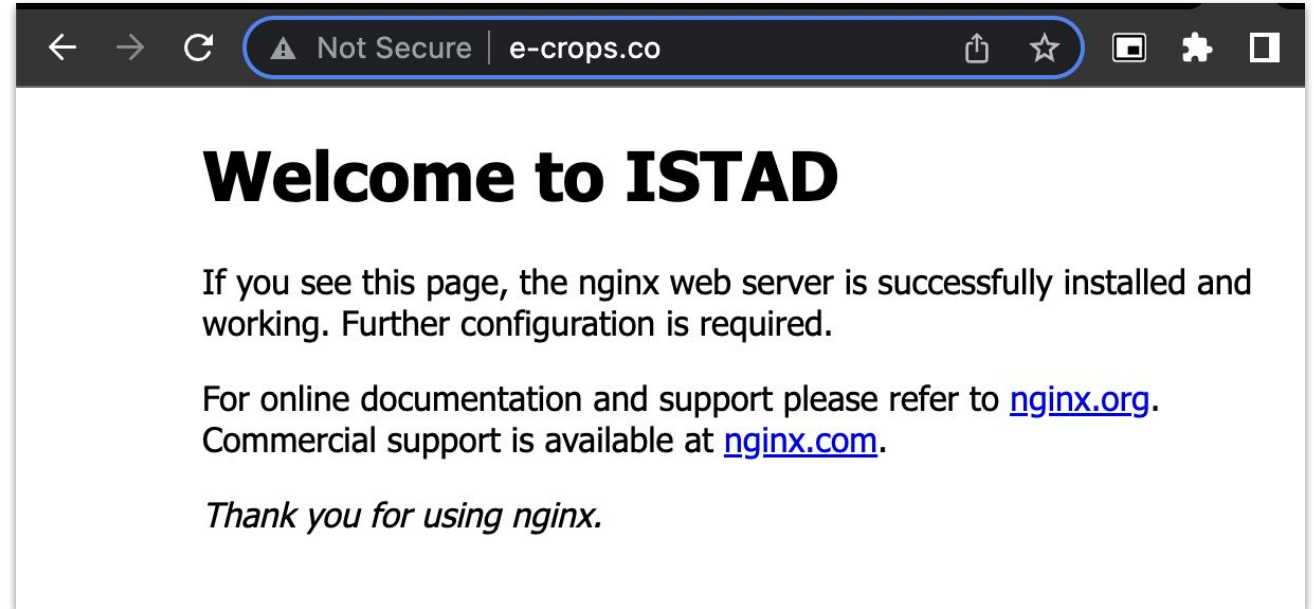
Nginx Web Server

- Config Domain Name in Namecheap

☐	Type	Host	Value	TTL	
☐	A Record	@	13.229.201.100	Automatic	
☐	CNAME Record	postgres	e-crops.co.	Automatic	
☐	CNAME Record	spring	e-crops.co.	Automatic	
☐	CNAME Record	www	e-crops.co.	Automatic	
 ADD NEW RECORD					

Nginx Web Server

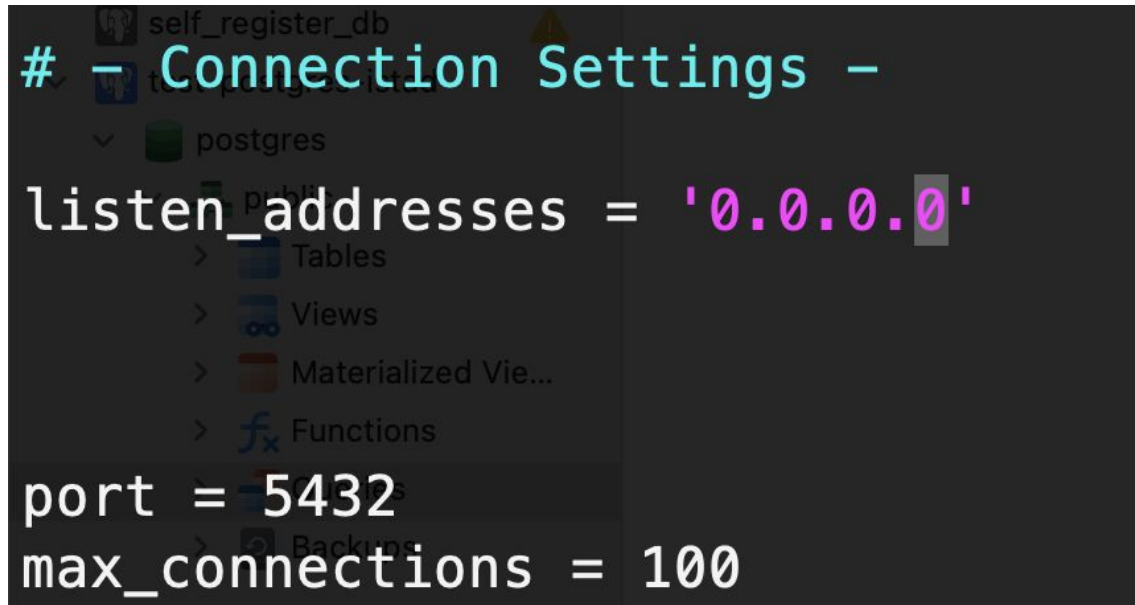
- `$ sudo ufw allow 22 #ssh`
- `$ sudo ufw enable`
- `$ sudo ufw allow 80`
- `$ sudo ufw reload`
- `$ sudo ufw status`



- `$ vim /var/www/html/index*.html # Edit file index*.html to change`
- `$ sudo nginx -t`
- `$ sudo systemctl daemon-reload`
- `$ sudo systemctl restart nginx`

Postgres with Nginx

- `$ sudo apt install postgresql postgresql-contrib -y`
- `$ cd /etc/postgresql/14/main`
- `$ sudo vim postgresql.conf`



```
# - Connection Settings -  
listen_addresses = '0.0.0.0'  
port = 5432  
max_connections = 100
```

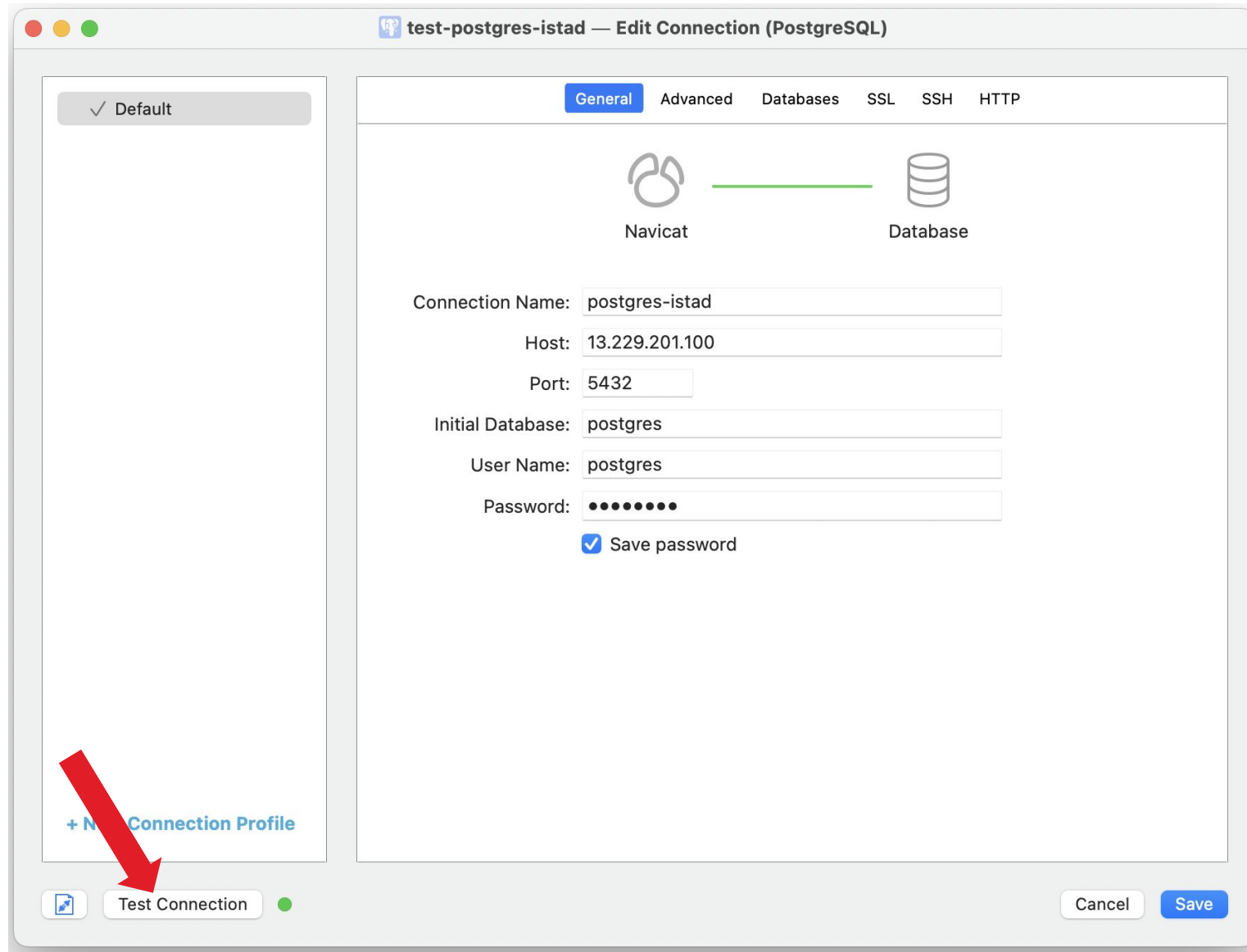
Postgres with Nginx

- `$ sudo vim pg_hba.conf`

```
local    replication    all                                peer
host     replication    all          127.0.0.1/32          scram-sha-256
host     replication    all          ::1/128              scram-sha-256
#Add this line to allow login with outside world
host     all            all          0.0.0.0/0            md5
```

- `$ sudo -u postgres psql`
- `#\password postgres`
 - Enter new password
- `$ sudo systemctl restart postgresql`
- `$ sudo systemctl enable postgresql`
- `$ sudo ufw allow 5432`
- `$ sudo ufw reload`

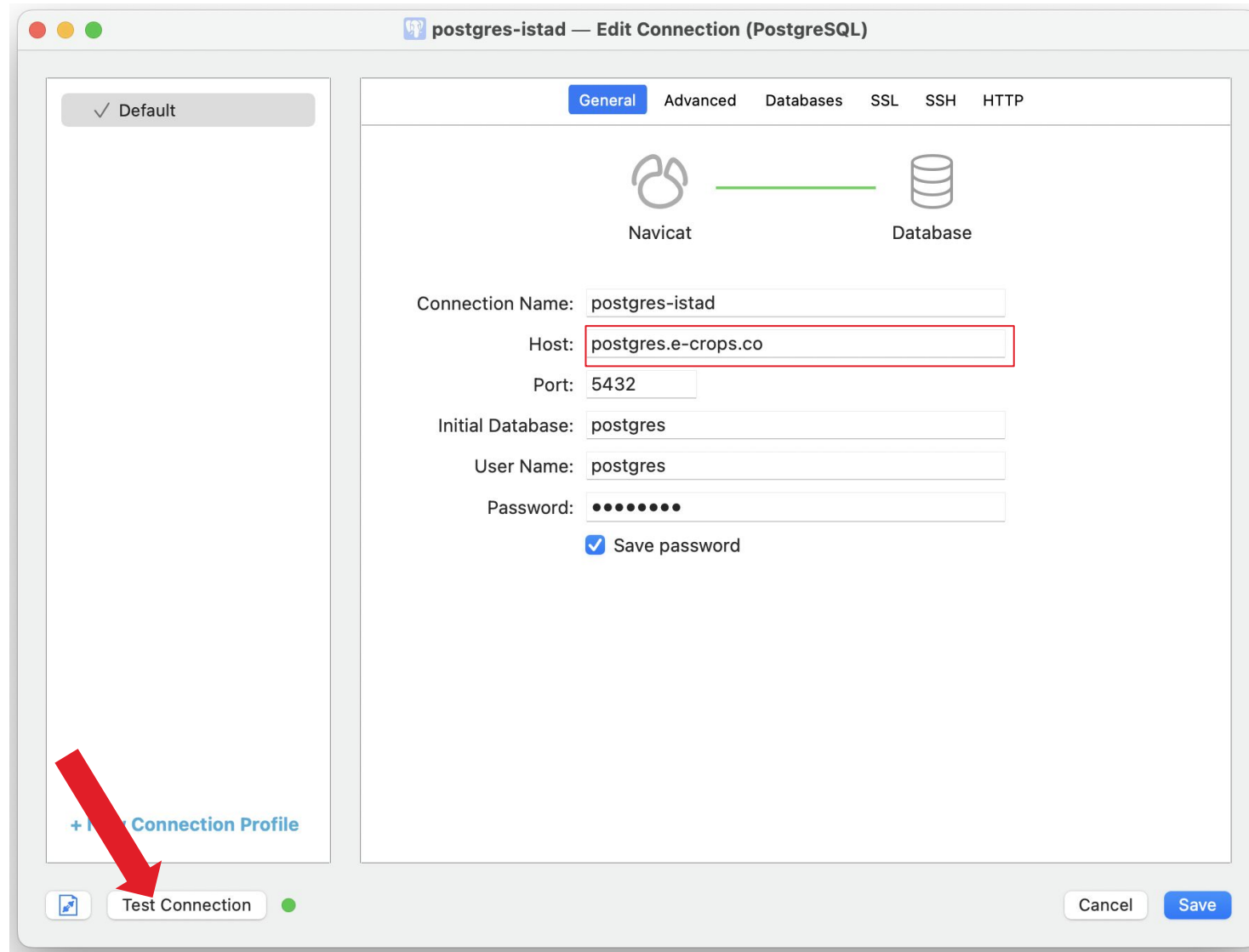
Postgres with Nginx



Postgres with Nginx

```
server {  
    listen 80;  
    listen [::]:80;  
    server_name postgres.e-crops.co;  
    location / {  
        proxy_pass http://0.0.0.0:5432;  
        proxy_set_header Host $http_host;  
        proxy_set_header X-Real-IP $remote_addr;  
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;  
        proxy_set_header X-Forwarded-Proto $scheme;  
  
        try_files $uri $uri/ =404;  
    }  
}
```

Postgres with Nginx



Spring boot with Nginx

- `$ mvn package` # Build project as *.jar file
- `$ scp registration-1.0.jar seng@13.229.201.100:/home/seng`
- # In Linux Server
- `$ sudo apt update`
- `$ sudo apt install openjdk-11-jdk`
- `$ sudo apt update`
- `$ sudo apt install maven`
- `$ java -jar registration-1.0.jar` # Run testing
- `$ sudo chmod +x registration-1.0.jar`
- `$ cd /etc/systemd/system/`
- `$ sudo vim spring-test.service`

Spring boot with Nginx

[Unit]

Description=SPRING-BOOT-TESTING

After=syslog.target

After=network.target

[Service]

User=seng

Type=simple

ExecStart=/usr/bin/java -jar /home/seng/registration-1.0.jar

Restart=on-failure

StandardOutput=syslog

StandardError=syslog

SyslogIdentifier=register

[Install]

WantedBy=multi-user.target

Spring boot with Nginx

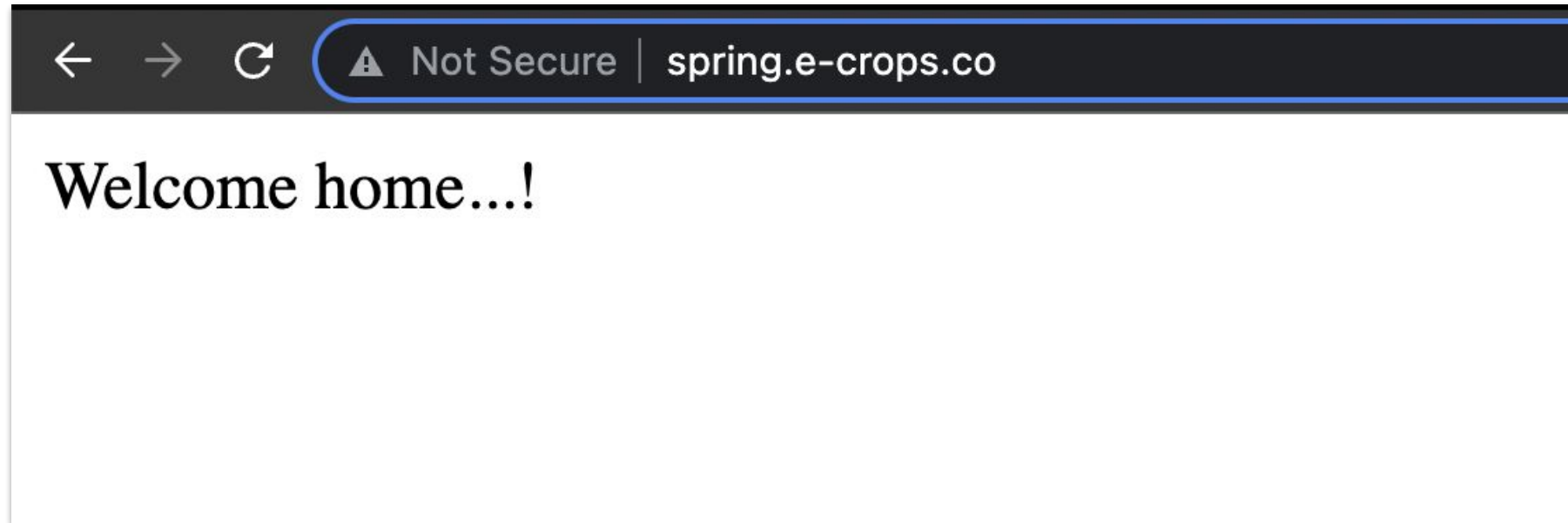
- `$ sudo systemctl start spring-test.service`
- `$ sudo systemctl enable spring-test.service`
- `$ sudo systemctl status spring-test.service`
- `$ curl localhost:8080 # Testing in local`
- `$ cd /etc/nginx/conf.d`
- `$ sudo vim spring-test.conf`

Spring boot with Nginx

```
server {  
    listen 80;  
    listen [::]:80;  
    server_name spring.e-crops.co;  
    location / {  
        proxy_pass http://0.0.0.0:8080;  
        proxy_set_header Host $http_host;  
        proxy_set_header X-Real-IP $remote_addr;  
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;  
        proxy_set_header X-Forwarded-Proto $scheme;  
  
        try_files $uri $uri/ =404;  
    }  
}
```

Spring boot with Nginx

- `$ sudo nginx -t`
- `$ sudo systemctl restart nginx`



Configure HTTPS Protocol

- **Let's Encrypt** is an open and automated certificate authority that uses the ACME (Automatic Certificate Management Environment) protocol to provide free TLS/SSL certificates to any compatible client. These certificates can be used to encrypt communication between your web server and your users.
- **Certbot** is by far the most popular Let's Encrypt client. It is included in most major Linux distributions, and includes convenient automatic configuration capabilities for Apache and Nginx.



Configure HTTPS Protocol

```
$ sudo apt update -y
```

```
$ sudo apt install certbot -y
```

```
$ sudo apt install python3-certbot-nginx -y
```

```
$ sudo nginx -t
```

```
$ sudo systemctl reload nginx
```

```
$ sudo certbot --nginx # For the first time or reconfiguration
```

```
$ sudo certbot --nginx -d example.com
```

```
$ sudo certbot renew --dry-run
```

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Thank you

Begin with the theory, Start with the practical